FINAL PRODUCT REPORT

World Radio

Architecture, functionality, user experience, security posture, quality verification, release readiness and operational handoff.

Version: 1.0.0+1

Framework: Flutter 3.47.1 / Dart 3.13.1

Report date: 25 August 2026

Repository: github.com/sahan8/world_radio

Assessment: Production candidate; store signing credentials remain an owner-controlled publishing gate.

1. Executive summary

World Radio is a cross-platform live-radio application that discovers stations through the public Radio Browser directory and provides background playback, search, discovery, favorites, history, theming, sleep timing and resilient playback controls.

The product has been taken from Flutter starter state to a cohesive release candidate with a responsive neumorphic interface, platform metadata, persistent local preferences, error handling, automated tests, and verified web, Windows and Android development artifacts.

> **Quality result:** Dart analysis reports no issues. All 14 automated tests pass. Hardened web and Windows release builds complete successfully, and the Android debug package builds successfully.

> **Publishing gate:** Android release packaging intentionally fails until the product owner supplies a private upload keystore. This prevents accidental distribution of a debug-signed release. iOS signing and a physical iOS build require Apple credentials and macOS/Xcode.

Release-readiness snapshot

Area	Status	Notes
Application code	PASS	57 Dart source files; 6,276 library lines.
Automated verification	PASS	14 tests across controller, widget, responsive, failure and security behavior.
Web release	PASS	Release bundle generated with deployable security-header policy.
Windows release	PASS	Release executable and runtime bundle generated.
Android functional package	PASS	Debug APK generated for device acceptance testing.
Android store bundle	OWNER GATE	Private keystore and key.properties required.
iOS store archive	OWNER GATE	Apple team, final bundle ID, certificates and macOS/Xcode required.

2. Product scope and user experience

Home

Featured live receiver, popular stations, search entry, country discovery, genre browsing and listening history.

Explore

Responsive country, genre and language directories with animated category switching.

Search

Debounced directory search, filters, quick suggestions, pagination, empty states and retry behavior.

Saved

Local favorites with immediate add/remove behavior and queue-aware station playback.

Player

Background audio, play/pause, previous/next, retry, volume, favorite, copy-link and real sleep timer.

Settings

System/light/dark appearance, listening-history management, privacy information and responsive layout.

Design system

The interface uses a monochrome receiver-inspired identity with reusable color, typography, radius, motion and neumorphic depth tokens. Raised and pressed surfaces use paired shadows and controlled contrast in both brightness modes. Interaction motion respects reduced-animation preferences.

- Maximum content widths preserve readability on tablets and desktop.
- Navigation remains reachable while the mini-player presents active playback.
- Custom controls include semantic labels and state descriptions.
- Approved mobile frames from 360 x 800 through 430 x 932 are regression-tested.
- Brightness and increased text scaling are included in widget tests.

3. Technical architecture

Layering

Layer	Responsibilities
Presentation	Flutter screens, reusable widgets, responsive layout, motion, accessibility and navigation.
Application	Player, home, search, library and theme controllers based on ChangeNotifier and inherited scopes.
Domain	Validated immutable radio-station model and display-derived properties.
Data	Radio Browser HTTP client, shared-preference storage, in-memory test fallback and audio handler.
Platform	audio_service, just_audio, audio_session, Android foreground media service, iOS background audio and web/desktop runners.

Primary data flow

1. The application requests a bounded station page from the HTTPS Radio Browser endpoint.
2. Response objects are parsed, deduplicated and rejected unless their stream URL passes strict public-HTTPS validation.
3. The player converts a selected station to a MediaItem and supplies the validated stream to just_audio.
4. Playback state is broadcast through audio_service and reflected in the mini-player, full player and platform notification.
5. Favorites, recent stations, last station and theme mode persist locally through asynchronous platform preferences.

Dependency baseline

Package	Version / constraint	Purpose
http	^1.6.0	Directory API requests
just_audio	0.10.6	Cross-platform stream playback
audio_service	0.18.19	Background media and system controls
audio_session	0.2.4	Audio-focus and noisy-device behavior
shared_preferences	2.5.5	Non-sensitive local preferences
flutter_lints	^6.0.0	Static quality rules

All direct dependencies were current during the final audit. OSV returned no known vulnerabilities across 77 resolved hosted packages.

4. Security and privacy assessment

Implemented controls

Control	Implementation
Transport security	Directory, station streams and artwork require HTTPS. Android cleartext traffic is disabled and the iOS arbitrary-media ATS exception was removed.
Destination validation	Rejects malformed URLs, embedded credentials, localhost, .local domains, loopback, RFC1918 private IPv4, link-local, multicast, private IPv6 and mapped private IPv4.
Secret management	.env, signing properties, keystores, certificates and private-key patterns are ignored. A tracked .env.example contains public configuration only.
Fail-closed signing	Android release tasks stop immediately when the ignored private signing configuration is absent.
Web policy	Baseline CSP, HSTS, no-sniff, no-frame, referrer and permissions headers are provided in web/_headers.
Minimal permissions	Android requests internet, wake-lock and foreground media playback permissions only.
Local privacy	No accounts, advertising SDK, analytics, location, camera, microphone or contacts. Favorites and history remain on device.
Logging	Errors contain station title and host but do not print credentials, tokens or full URLs.

Audit evidence

- No API keys, passwords, tokens, private keys, certificates or keystores were found in tracked files or Git history.
- The Radio Browser endpoint and user-agent are public client configuration, not credentials.
- The exported Android audio service and media receiver match audio_service integration requirements.
- shared_preferences stores only non-critical station selections and appearance settings.
- Security validation is independently exercised by four focused SafeUri test groups.

> **Residual limitation:** Client-side host checks cannot fully defeat DNS rebinding after hostname resolution. Stronger enforcement would require a controlled backend proxy that resolves and filters destinations before relaying streams. This product deliberately avoids a backend and accepts that bounded residual risk.

5. Reliability, accessibility and testing

Reliability behavior

- Directory requests use bounded page sizes and 12-second timeouts.
- Search debouncing and request identifiers prevent stale responses from overwriting current results.
- Controllers close HTTP clients, timers and stream subscriptions on disposal.
- Audio preparation times out after 15 seconds and presents retry/next-station recovery.
- Storage corruption is skipped or cleared without preventing application startup.
- Sleep timer cancellation and disposal prevent orphaned timers.

Automated test inventory

Category	Coverage
Library	Favorite persistence/deduplication and bounded recent-history ordering.
Navigation	Every primary destination opens and remains accessible.
Responsive UI	Home scrolling, Explore switching and five approved mobile frame sizes.
Playback failure	Invalid stream produces recoverable mini/full player error state.
Accessibility	Interactive UI remains visible across light/dark modes and 1.4x text scale.
Settings	Narrow-screen rendering and theme switching without Material or overflow exceptions.
Timer	Sleep timer expires, pauses through the handler and resets its state.
URL security	Public HTTPS acceptance and rejection of HTTP, credentials, local/private, malformed and unsupported targets.

> **Final automated result:** 14 tests passed; Dart analyzer reported no issues; git diff whitespace validation passed.

6. Build and deployment report

Target	Result	Artifact / requirement
Web release	BUILT	build/web; includes _headers policy. Host exclusively over HTTPS.
Windows x64 release	BUILT	build/windows/x64/runner/Release; distribute the complete Release directory.
Android debug	BUILT	build/app/outputs/flutter-apk/app-debug.apk; device testing only.
Android release	BLOCKED SAFELY	Requires owner upload keystore. Obsolete debug-signed AAB was removed.
iOS release	OWNER GATE	Requires Apple Developer team, final bundle identifier, distribution profile and macOS/Xcode.
macOS release	PLATFORM GATE	Requires macOS build host and network-client entitlement review.

Owner action: Android store signing

Create and securely back up a private upload key, then create ignored `android/key.properties`:

```
storeFile=C:/secure/location/world-radio-upload.jks
storePassword=OWNER_SECRET
keyAlias=world-radio-upload
keyPassword=OWNER_SECRET
```

Then build:

```
flutter build appbundle --release --dart-define-from-file=.env
```

Never commit the keystore or key.properties. Store encrypted backups in at least two owner-controlled locations.

Owner action: web deployment

1. Deploy the entire build/web directory to an HTTPS-only static host.
2. Confirm the host applies web/_headers; configure equivalent headers manually if unsupported.
3. Verify CSP in browser developer tools and test station playback from the production origin.
4. Enable immutable caching for hashed assets while keeping index.html revalidated.

7. Configuration and operations

Local configuration

The ignored .env file contains public build configuration:

```
RADIO_API_BASE_URL=https://de1.api.radio-browser.info
RADIO_API_USER_AGENT=WorldRadio/1.0 Flutter
```

Run and build with `--dart-define-from-file=.env`. Values compiled into Flutter clients are observable and must never contain private credentials.

Standard verification commands

```
flutter pub get
dart analyze
flutter test --no-pub --dart-define-from-file=.env
flutter build web --release --no-pub --dart-define-from-file=.env
flutter build windows --release --no-pub --dart-define-from-file=.env
flutter build appbundle --release --no-pub --dart-define-from-file=.env
```

Monitoring recommendations

- Run analysis and tests on every pull request.
- Run dependency-outdated and OSV checks on a scheduled basis.
- Track directory request failures, stream preparation failures and crash-free sessions without logging station URLs or user history.
- Test current Android/iOS releases after Flutter, audio_service or just_audio upgrades.
- Rotate store credentials only through the platform-supported process and preserve historical signing continuity.

Known product tradeoff

Strict HTTPS improves confidentiality and integrity but excludes legacy stations that publish only HTTP streams. This is intentional for the final secure baseline. If compatibility becomes a business requirement, introduce a controlled HTTPS relay rather than re-enabling arbitrary cleartext traffic in the client.

8. Final acceptance checklist

Item	State
Product features implemented and integrated	COMPLETE
Neumorphic responsive UI across primary experiences	COMPLETE
Runtime Material ancestry defects resolved	COMPLETE
HTTPS-only and public-destination validation	COMPLETE
Secret patterns and Git history reviewed	COMPLETE
Dependency currency and advisory scan	COMPLETE
Analysis and automated tests	COMPLETE
Web, Windows and Android device-test artifacts	COMPLETE
Private Android upload signing	OWNER REQUIRED
Apple signing and physical-device acceptance	OWNER REQUIRED
Store listings, privacy URL and screenshots	OWNER REQUIRED

> **Conclusion:** World Radio is a tested, hardened production candidate. No known code, dependency or repository-secret issue blocks controlled beta testing. Public-store delivery is intentionally gated only by private owner credentials, platform account setup and final physical-device acceptance.

Reference guidance

- Android cleartext communication risks
- Android Network Security Configuration
- Apple App Transport Security
- audio_service platform setup
- shared_preferences storage characteristics
- Open Source Vulnerabilities database